

Part 1:

1. Raw Inputs The raw input is a 3-axis accelerometer (accX, accY, accZ) sampled at 100 Hz.

Data was collected in 3-second windows under two conditions: fan running normally (nominal) and fan powered off (off).

2. NN Classifier Features The NN classifier uses **spectral features** produced by a Spectral Analysis DSP block (FFT-based with a high-pass filter), with **33 input features** total. Five actual feature names are: accX RMS, accX Peak 1 Height, accY Spectral Power 2.0–5.0, accZ Spectral Power 2.0–5.0, and accX Peak 1 Frequency.

3. NN Classifier Outputs The classifier outputs probability scores for **2 classes: nominal and off**, using a softmax activation on the output layer. It achieved 100% accuracy on the validation set.

4. Anomaly Detector Features The anomaly detector uses **spectral features** selected by Edge Impulse's Feature Importance tool, with **4 features** chosen from the 33 available. The feature names are: accX RMS, accX Peak 1 Height, accY Spectral Power 2.0–5.0, and accZ Spectral Power 2.0–5.0.

5. Deployment Evidence

Model optimizations and performance

Model optimizations can increase on-device performance but may reduce accuracy.
Performance estimate for Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz).

Quantized (int8)
Selected ✓

	SPECTRAL FEATUR...	NN CLASSIFIER	TOTAL
LATENCY	16 ms.	1 ms.	17 ms.
RAM	7.1K	-	7.1K
FLASH	-	-	-
ACCURACY			-

Unoptimized (float32)
Select

	SPECTRAL FEATUR...	NN CLASSIFIER	TOTAL
LATENCY	16 ms.	1 ms.	17 ms.
RAM	7.1K	-	7.1K
FLASH	-	-	-
ACCURACY			-

Build output

Backing up deployment OK

Download URL: <https://studio.edgeimpulse.com/v1/api/989480/deployment/history/11/download>

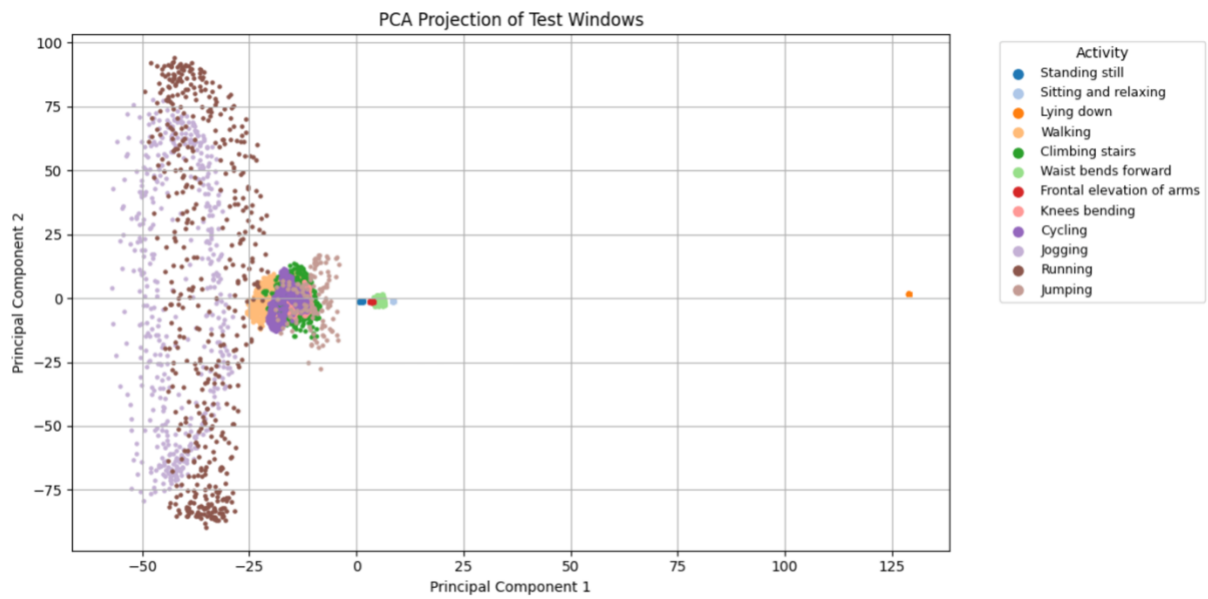
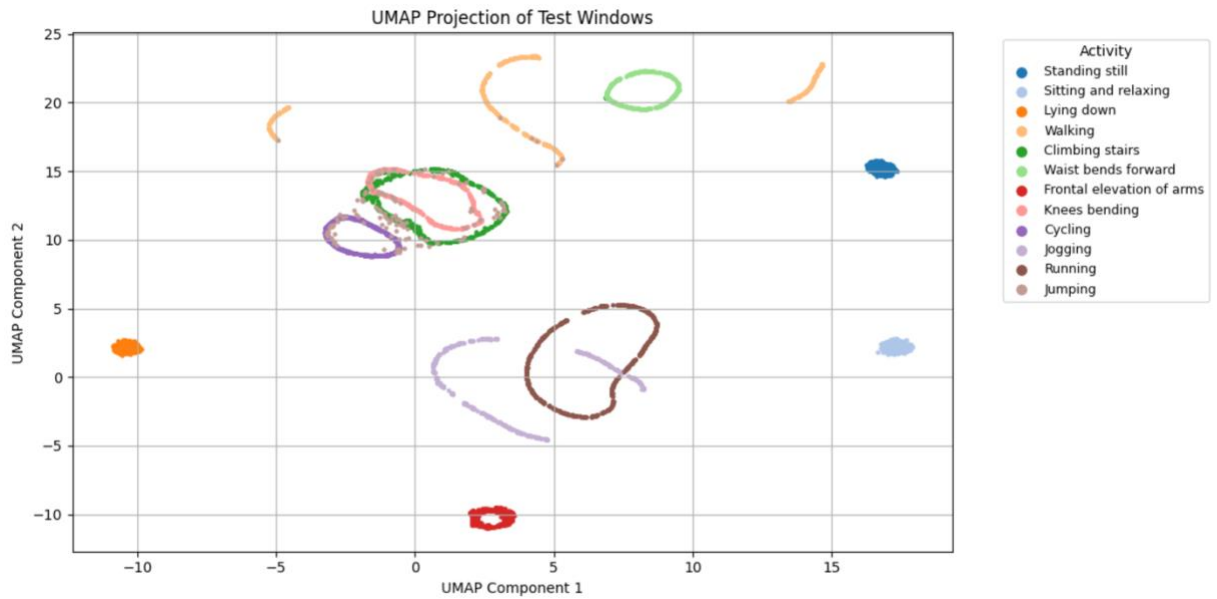
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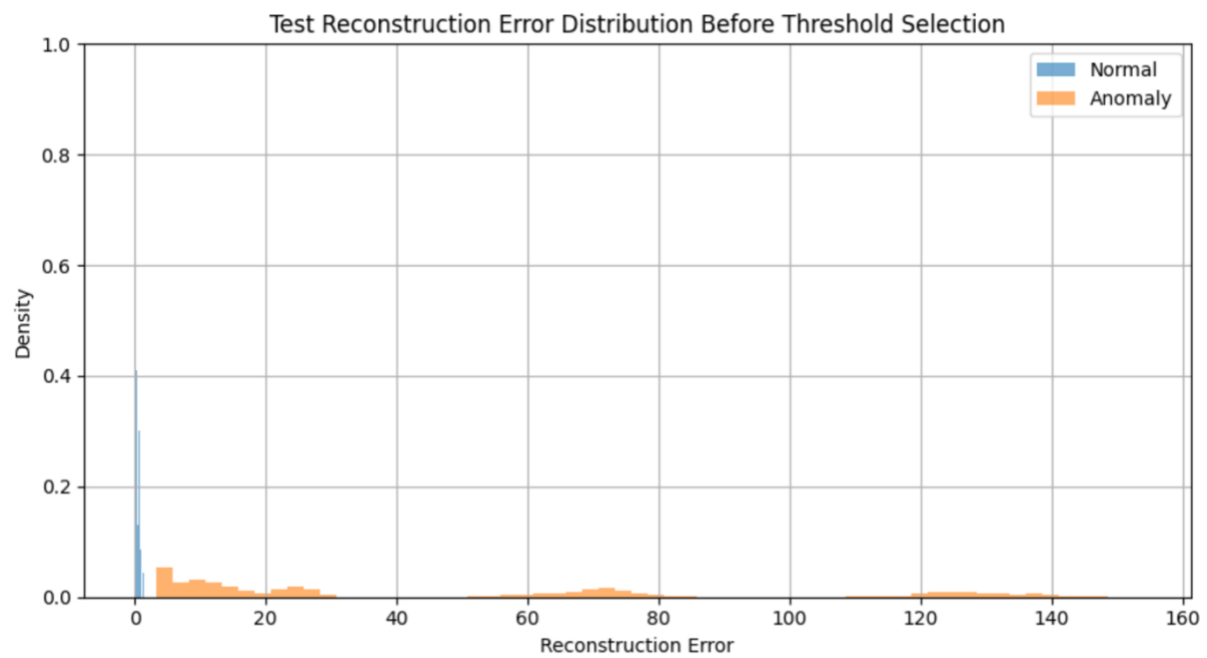
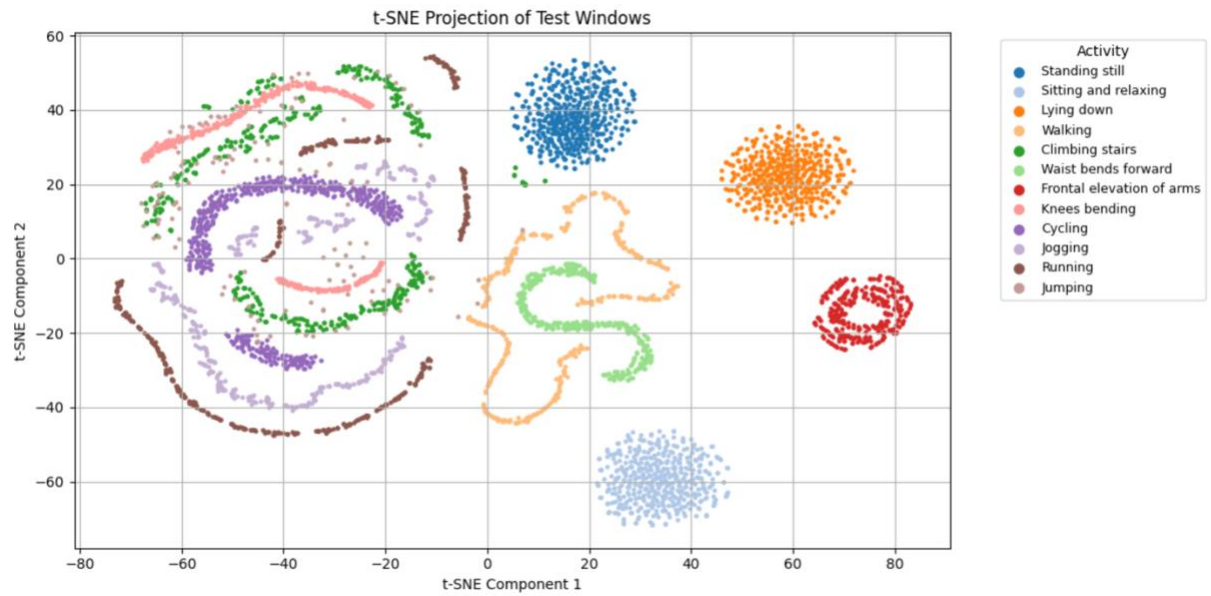
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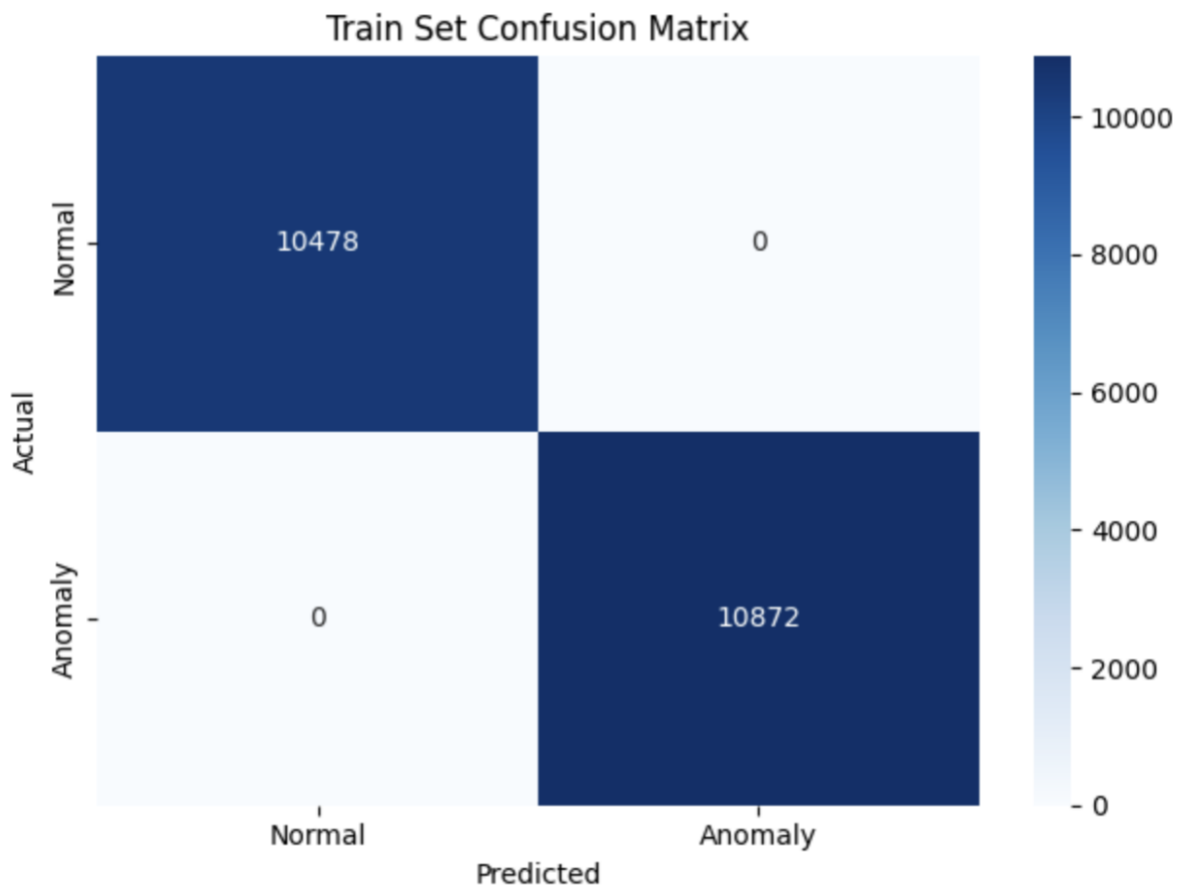
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Job completed (success)
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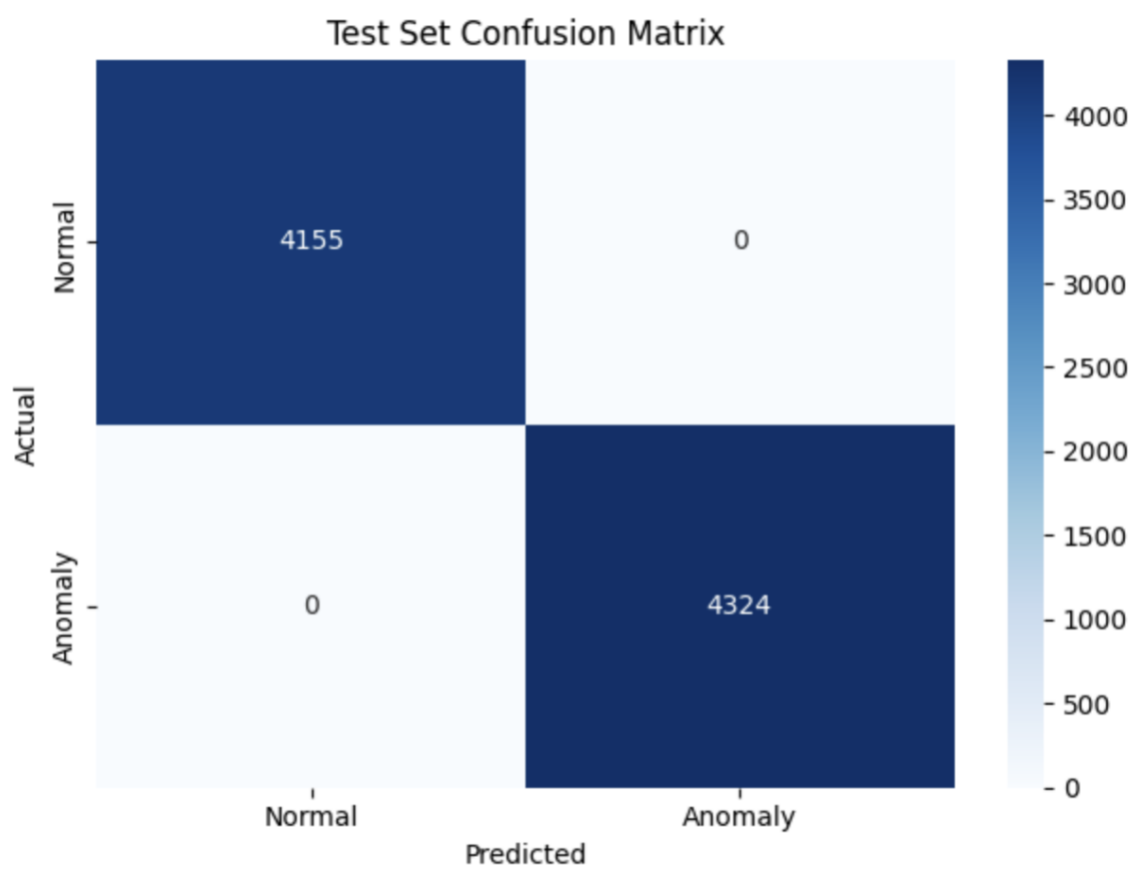
6. Short Interpretation The classifier will always output one of its two trained labels (nominal or off), even for genuinely abnormal inputs, so it cannot be fully trusted in isolation. The anomaly score is the more reliable signal — a high score indicates the input falls outside the learned distribution, even if the classifier appears confident.

Part 2:

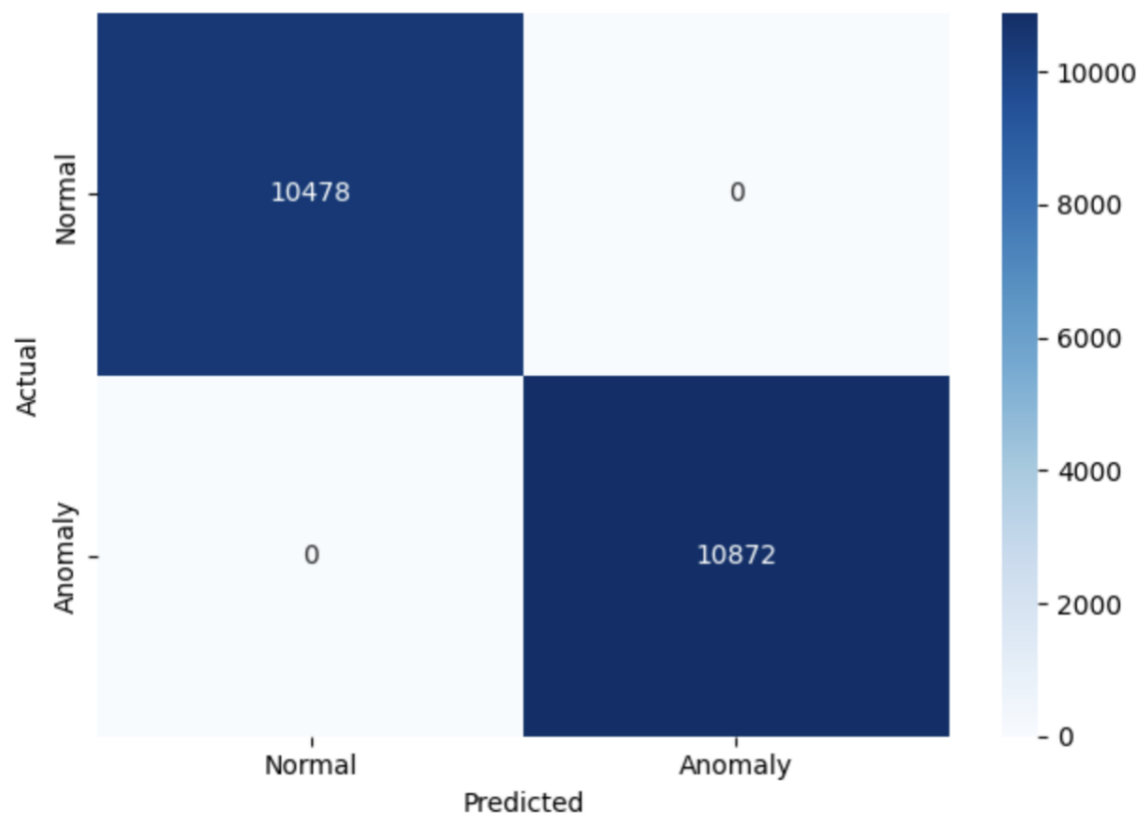




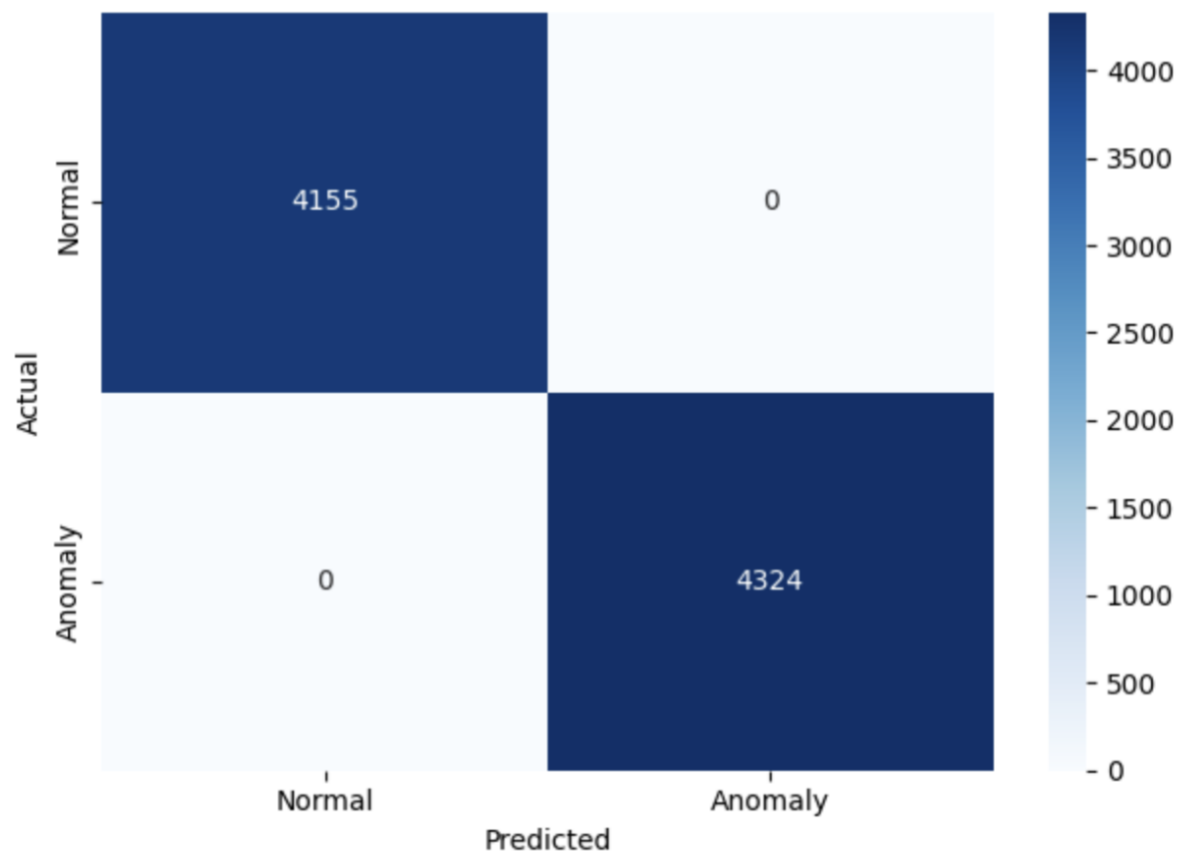




Train Set Confusion Matrix: Int8 Model



Test Set Confusion Matrix: Int8 Model



Train set evaluation: int8 model				
	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	10478
Anomaly (1)	1.00	1.00	1.00	10872
accuracy			1.00	21350
macro avg	1.00	1.00	1.00	21350
weighted avg	1.00	1.00	1.00	21350
Test set evaluation: int8 model				
	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	4155
Anomaly (1)	1.00	1.00	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479

Anomaly Detection Threshold A threshold around 2 is typical, chosen by observing that normal samples produce anomaly scores higher than 2 during training.

Reconstruction Error Normal data reconstructs with low error since the model learned its patterns, while anomalous data differs from the training distribution and reconstructs poorly, yielding high error. This difference in reconstruction error is what flags a sample as anomalous.

Information to Preserve from the Notebook The trained model weights, feature scaling parameters (mean and standard deviation), and input feature order must all be preserved when deploying. Without these, the model receives incorrectly formatted inputs and produces meaningless results.

Why Preprocessing Must Match The model was trained on data filtered, windowed, and scaled in a specific way, so any deviation on the Arduino will produce input features outside the expected distribution. This mismatch causes misclassification or unreliable anomaly scores even on normal data.

Main Limitations on a Microcontroller The model only learns normal states, so novel-but-normal conditions (e.g., different fan speed) can be falsely flagged as anomalies. Limited RAM and compute also restrict model complexity, window size, and the number of usable features.